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RECEIVED 13 November 2025
REVISED 23 March 2026
ACCEPTED 23 March 2026
PUBLISHED 14 April 2026

CITATION
Caspari-Sadeghi S (2026) AI literacy for
teacher educators: a holistic curriculum
for capacity-building in higher
education.
Front. Educ. 11:1745768.
doi: 10.3389/feduc.2026.1745768

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AI literacy for teacher educators: a holistic curriculum for capacity- building in higher education

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The emerge of Generative Artificial Intelligence (GenAI) sparked diverse reactions within the educational sector, from enthusiasm to prohibition. The core question is no longer whether educators should use GenAI, but rather what skills, knowledge, and mindset they need to use AI effectively and responsibly. Educational AI literacy focuses on the competencies educators must develop to understand, utilize, and critically evaluate AI in teaching, learning, and assessment. However, fostering educational AI literacy requires more than simply identifying and defining necessary competencies; existing literature does not adequately address providing a comprehensive curriculum for developing AI literacy in educational contexts. Drawing on an integration of conceptual theory, empirical research, and the author's extensive experience in faculty development across diverse international contexts, this conceptual paper introduces a holistic, participatory, and inquiry-driven curriculum framework for developing AI literacy among teacher educators in higher education. The synergy of these three sources of knowledge ensures a theoretically grounded and practically relevant foundation for the framework's design. Developed through an iterative process of design, implementation, and evaluation, the framework offers structured guidance for creating, embedding, and assessing AI literacy initiatives responsive to the unique needs and institutional contexts of higher education settings. Rather than prescribing a fixed hierarchical sequence, the framework is intentionally designed as a flexible blueprint—an open-ended roadmap, where the curriculum's depth and breadth can be adapted to an institution's AI-readiness and its broader mission regarding educators' expected AI proficiency or mastery. Additionally, the analytical study addresses potential implementation challenges, strategies to mitigate them, and implications for stakeholders.

KEYWORDS

AI curriculum, AI literacy for teacher educators, AI readiness, participatory design, utilization-focused evaluation

1 Introduction: AI in education

Digitalization and the advancement of technologies such as Artificial Intelligence (AI) have transformed how we live, learn, and communicate. AI can broadly be defined as a set of computational algorithms capable of performing tasks typically associated with human intelligence, such as pattern recognition, perception, decision-making, learning, and language use (Banh and Strobel, 2023; Winston, 1993). A more functional definition—one that avoids direct comparison with human cognition—frames AI as “automation based on associations,” wherein computers automate reasoning by identifying patterns in data or deriving associations from expert knowledge (U.S. Department of Education, 2023).

Artificial intelligence in Education (AIED) emerged over 60 years ago, with its primary goals including personalizing learning, enhancing learners' engagement and retention, automating routine teaching tasks, and streamlining administrative processes. AI-driven technologies—such as intelligent tutoring systems, adaptive learning environments, automated scoring tools, real-time feedback mechanism, learning analytics, and educational data mining—have been widely implemented in educational practices (Caspari-Sadeghi, 2023a, b; Neshaei et al., 2024).

More recently, breakthroughs in Generative Artificial Intelligence (GenAI), particularly in Large Language Models (LLMs), have revolutionized the field of natural language processing (NLP)—a domain of AI focused on understanding and generating human language. AI-powered tools such as ChatGPT, Claude, Dall-E, and Synthesia can mimic human creativity, answer questions, solve mathematical problems, write essays, explain complex concepts, transcribe audio, and produce multimedia content (World Economic Forum, 2024; Yan et al., 2023). The application of LLM in education has produced mixed outcomes, both positive (e.g., enhanced self-regulated learning) and negative (e.g., over-reliance and dependency), depending on the employed learning and pedagogical approach (Bastani et al., 2024; Kasneci et al., 2023; Prihar et al., 2023). Teachers thus play a pivotal mediating role in optimizing the benefits of GenAI while mitigating its potential harms and risks. To harnesses the advantages of GenAI effectively, educators must be equipped with the necessary knowledge and competencies to integrate these technologies in educational settings. Additionally, teachers should be informed of the ethical risks associated with AI-driven educational technologies to ensure their responsible and safe use.

Despite the growing awareness about the importance of AI in education, particularly for teachers and teacher educators, current research and policy initiatives remain underdeveloped in several key areas. First there is a lack of comprehensive, empirically grounded frameworks tailored specifically to the professional development needs of teacher educators in higher education. Second, the current literature pays limited attention to the factors that influence the success and sustainability of such educational initiatives—such as institutional AI readiness, technological infrastructure, human resource capacity, and organizational culture. Third, most existing professional development efforts tend to focus narrowly on the technical mastery of AI tools—often through sandbox environments—without adequately addressing their pedagogical integration or alignment with core educational principles.

This analytical paper aims to address these gaps by proposing a structured yet flexible framework for embedding AI literacy for teacher educators within higher education. The objective is to provide a practical and adaptable model that supports the systematic and sustainable development of AI literacy in educators. Drawing on relevant literature (Ng et al.'s AI literacy framework, 2021), empirical work (e.g., inquiry-based approach to professional development), and the author's extensive experience—including initiatives focused on teacher assessment literacy, data literacy, and digital competence curricula in international contexts—we propose a framework which highlights sustainability, inclusivity, and co-ownership.

The conceptual study is organized as follows: first, we examine the need for AI literacy in education, reviewing existing theoretical frameworks, conceptualizations, programs and initiatives designed to foster AI literacy. Next, we introduce a holistic, participatory, and inquiry-based approach to designing, implementing, and evaluating a comprehensive AI literacy curriculum tailored to teacher educators. Each phase of the curriculum design is discussed in detail. Finally, we identify key challenges associated with implementing the proposed curriculum and suggest strategies to address them. We conclude with recommendations for future research and practical implications for stakeholders.

2 The need for AI literacy in higher education

“Without universal AI literacy, AI will fail us.”

—World Economic Forum, 2022.

Traditionally, literacy has been defined as the ability to access, share, and communicate ideas through written language. However, advancements in science, technology, and society expanded the scope of essential literacies required to survive and thrive in rapidly evolving complex world. These now include digital literacy, computational literacy, data literacy, media literacy, information literacy, and digital citizenship (Jacob and Warschauer, 2018; Leu et al., 2013; Njenga, 2018).

Among these, digital and data literacy are fundamental to AI literacy. Digital literacy—the ability to use ICT and digital technologies such as the internet, social media, and mobile for communication, learning, working, or completing a task—is a prerequisite for developing AI literacy (Chiu et al., 2024; Falloon, 2020). Moreover, data literacy can be a significant facilitator of educational AI literacy due to the centrality of data in machine learning (a subfield of AI). It requires the educators to understand, use, and reflect on data-informed decision-making process, e.g., data collection, analysis, visualization, and interpretation, to improve educational outcomes (Mandinach and Gummer, 2016).

Although AI literacy builds on all previous literacies, it goes beyond them due to the unique capacity of AI to mimic human intelligence. Furthermore, the current AI technologies have distinctive features that differentiate them from previous generations of human-made technologies, including (a) autonomous decision-making, (b) self-learning capability, and (c) inscrutability—the inherent lack of explainability and interpretability often associated with neural network architectures and deep learning models— (Berente et al., 2021; Pinski and Benlian, 2024).

AI literacy is a critical competency that must be urgently developed across all sectors of society, including among public and professions. Technological literacy (e.g., AI and big data) is among employers' top priorities for 2027. It is predicted that by 2030, over 40% of the core skills required for all jobs will change due to technological advancements (World Economic Forum, 2024). The urgency of this transformation is reflected in the UNESCO Sustainable Development Goals (SDGs) which

emphasizes education’s role in eradicating poverty and fostering sustainable growth through inclusive, equitable, and lifelong learning opportunities for all. However, the main barrier is the global shortage of qualified teachers capable of preparing students to live, work, and learn in an AI-rich world. This underscores the pressing need for higher education institutions to develop and implement comprehensive curricula that foster AI literacy among both educators and students.

3 AI literacy: theoretical frameworks and programs

Conceptualizing AI literacy is a critical step in guiding curriculum development, instructional design, and assessment. Theoretical definitions and specification of proficiency dimensions directly shape how educators are trained to engage with AI technologies. The first discussion on AI Literacy emerged a decade ago, suggesting that users should have basic knowledge and ability to interact with AI-based platforms, services, and products (Kandlhofer et al., 2016; Konishi, 2015). Since then, interest in defining the components of AI literacy has grown significantly. This section reviews key definitions and theoretical frameworks that outline the core dimensions of AI literacy.

In their pioneering work, Long and Magerko (2020) conceptualized AI literacy based on *five big ideas about AI* (Touretzky et al., 2019), which describes how machines or computers perceive and interact with their surrounding environments:

- Perception: AI systems perceive the world through sensors (e.g., computer vision).
- Interaction: AI communicate using natural language (e.g., natural language processing).
- Representation: AI models, represents, and reasons about its surrounding world.
- Learning: AI learns and improves automatically from data (e.g., machine learning)
- Impact: AI influences society, both positively and negatively (e.g., ethical considerations).

Building on these principles, Long and Magerko proposed AI literacy consists of 17 competencies, which can be categorized into three broad areas: (1) understanding AI concept, (2) using AI applications, and (3) complying with ethical guidelines. Several design activities were suggested to develop these competencies among educators.

Ng et al. (2021) adopted a revised version of Bloom’s Taxonomy to define AI Literacy components or dimensions. Bloom’s taxonomy (1956) classifies cognitive skills into six hierarchical levels—remember, understand, apply, analyze, evaluate, and create— which has long been used to guide instructional design and learning progression across diverse contexts. Based on this taxonomy, they proposed four interdependent dimensions of AI literacy:

- (a) Know and understand AI: basic comprehension of AI concepts and functions.
- (b) Use and apply AI: Practical utilization of AI tools.

- (c) Evaluate and create AI: critical assessment of AI systems and development of new AI solutions.

Additionally, (d) AI ethics is also proposed as the fourth category which should be integrated in AI literacy programs. In this conceptual paper, we use the theoretical framework suggested by Ng et al. (2021)

Mills et al. (2024) similarly defined AI literacy as the knowledge and skill required to (a) understand, (b) use, and (c) evaluate AI safely, ethically, and effectively in both personal and professional contexts. Their framework highlights the centrality of *human judgement* and *justice* as core values across all components of AI literacy. Furthermore, they distinguish between different modes of engagement with AI: interacting with AI in daily life, using AI to perform job-related tasks, or developing AI systems.

The UNESCO AI Competency Framework for Teachers further advances this conceptual landscape by defining five core dimensions of AI literacy: (a) AI foundations and applications, (b) AI pedagogy, (c) AI ethics, (d) AI for professional development, and (e) human-centred mindset (Miao and Cukurova, 2024). Each dimension has three mastery levels—acquire, deepen, and create— to ensure all teachers, regardless of their disciplinary background or prior knowledge, can progressively develop a foundational level of AI competency for educational practices. Table 1 presents a concise summary of key AI literacy frameworks, highlighting their core components and focus areas.

Taken together, these frameworks consistently position AI literacy as a multidimensional construct, that extend beyond technical knowledge of AI tools to include critical assessment and ethical considerations. While we may use AI literacy and AI competency interchangeably, we acknowledge Chiu’s (2025) distinction that AI literacy refers to understanding what AI does and its limitations, whereas AI competency involves the ability to apply AI effectively. Literacy provides the essential

TABLE 1 Summary of AI literacy frameworks.

Framework	Conceptual basis	Dimensions/ Components
Long and Magerko (2020)	Five Big Ideas of AI	Perception; Interaction; Representation; Learning; Impact
Ng et al. (2021)	Revised Bloom’s Taxonomy	Know & understand AI; Use & apply AI; Evaluate & create AI; AI ethics
Mills et al. (2024)	Human-centred and justice-oriented approach	Understand AI; Use AI; Evaluate AI
UNESCO AI Competency Framework for Teachers (Miao and Cukurova, 2024)	Teacher professional competency model	AI foundations & applications; AI pedagogy; AI ethics; AI for professional development; Human-centred mindset

foundation, but competency builds on this knowledge to enable skillful, ethical, and productive use of AI.

Beyond theoretical conceptualizations, numerous institutional, national and international programs have been developed to promote AI literacy in education. Several countries, including China, USA, Australia, South Korea, Japan, and Singapore, have pioneered comprehensive policies and programs to develop AI literacy in education, particularly at K-12 levels.

For example, the AI for Teachers (*AI4T*) program, an EU initiative delivered through Massive Open Online Courses (MOOCs), consists of four modules: (1) introduction to AI in Education, (2) foundational AI knowledge, (3) applications and technical aspects, and (4) AI ethics. It allows teachers to learn about AI, access and explore AI tools and platforms, experiment with AI applications such as prompt engineering in ChatGPT, and integrate AI in their daily teaching practices. Other notable programs include *AI4K12*, *AI for kids* (*AI4K*), and *AI for Students* (Heintz, 2021; Laupichler et al., 2022). At the higher education level, institutions like the University of Florida have introduced initiatives such as *AI Across the Curriculum*, which aim to equip undergraduate students with AI competencies across all disciplines (Southworth et al., 2023).

All these initiatives and programs highlight the world-wide recognition of AI literacy as an essential competency for educational stakeholders such as teachers and students. Although AI literacy has begun to appear in several national curricula, Sperling et al. (2024) suggest that efforts to systematically integrate it into teacher education programmes are still limited.

While the above frameworks provide foundations for conceptualizing AI literacy, several gaps remain. First, many existing frameworks, e.g., *AI4K12*, are developed for K-12 education, focusing on schoolteachers and students. Consequently, there is limited conceptual guidance specifically addressing higher education, where AI literacy must also support disciplinary learning and research practices. Second, existing frameworks vary in their focus: some focus mainly on technical understanding of AI systems (e.g., Long and Magerko, 2020), while others highlight ethical, pedagogical, or human-centered dimensions (e.g., Miao and Cukurova, 2024; Mills et al., 2024). This variation reflects an ongoing debate about whether AI literacy should prioritize technical competence, critical understanding, or responsible use in real-world contexts.

These limitations highlight the need for broader conceptual perspectives that extend beyond K-12-oriented frameworks and support the systematic and holistic development of AI literacy in higher education. In response to this gap, the present conceptual article proposes an integrative framework designed to guide the coordinated development of AI literacy across institutional, organizational, and individual levels within universities.

4 AI literacy curriculum for teacher educators

We propose a framework for integrating AI literacy into higher education which is grounded in a triangulated foundation to ensure both theoretical rigor and practical relevance. First, it draws theoretical framework developed by Ng et al. (2021), which was discussed in previous section.

Second, it is informed by empirical research on instructional design for faculty development, capturing key challenges and success factors in implementing pedagogical innovation. Third, it is deeply rooted in the author's decade-long experience leading faculty development programs across a range of international contexts—teacher assessment literacy in Norway, data literacy and digital literacy initiatives in Germany, and evidence-based teaching reform in Iran. This experiential dimension allows the framework to be sensitive to institutional diversity, cultural variation, and systemic constraints.

We propose a *participatory*, *inquiry-based*, and *system-wide* curriculum for integrating AI literacy programs into higher education. This framework serves as an open-ended roadmap or flexible blueprint rather than a prescriptive, hierarchical method. It fosters local agency, cross-disciplinary collaboration, and iterative refinement, making it particularly suited for the dynamic context of higher education.

By participatory, we emphasize the active involvement of key stakeholders—including teacher educators, institutional leaders, instructional designers, and module organizers—throughout the co-design process. Engaging these actors in all phases fosters a deep understanding of the process, increases ownership, and enhances commitment to utilization of resultant findings (Dilek et al., 2025; Patton and Campbell-Patton, 2022). By inquiry-based, we ensure that judgement and decisions regarding program strengths, weaknesses, revisions, improvements, or restructuring are grounded in relevant evidence and systematically collected data. Rooted in constructivist learning theory, inquiry-based learning encourages educators to engage in questioning, exploration, experimentation, and reflection. It supports deeper learning by positioning teacher educators as active co-creators of knowledge (Dobber et al., 2017).

The proposed *AI literacy curriculum for teacher educators* consists of three interconnected phases—design, implementation, and evaluation—each guided by participatory and inquiry-based principles. These phases reflect the core components of a Design-Based Program, which enables iterative refinement through continuous evidence gathering and critical reflection. Grounded in Design-Based Research (DBR) methodology, this approach integrates theory and practice in real-world educational contexts, allowing for the systematic testing and improvement of pedagogical models. By engaging participants as co-designers and reflective practitioners, the process fosters both contextual relevance and theoretical advancement, ensuring that the curriculum evolves responsively in line with empirical insights and user feedback.

It is important to underscore that the primary goal of our proposed AI literacy program is not to train educators to become AI experts or talents who can code, program, develop, or optimize AI algorithms, models, and solutions. Rather, the training program aims at equipping educators with knowledge and skills necessary to confidently, effectively, and responsibly integrate AI into their teaching and professional practice.

4.1 A. Design phase

The design phase of the AI literacy curriculum involves two key steps: (a) assessing *AI readiness* and (b) defining core AI competencies for teacher educators.

AI readiness refers to the technological preparedness of both individual educators and educational institutions to adopt and integrate AI technologies (Martinez Plumed, 2020). A baseline assessment is essential to identify existing gaps, deficiencies, needs, and available resources to inform program's development. Key dimensions of AI readiness include:

Institutional readiness: this encompasses cultural and policy aspects, such as institution's openness to innovation, leadership strategies that fosters an inclusive environment, where everyone regardless of their expertise or experiences are welcome to join and contribute, and the presence (or absence) of institutional policy or ethical regulations on AI. Furthermore, previous projects on digitalization as well as the current level of digitalization in administration, teaching, and research level are also crucial. For example, the University of California had a strong AI culture prior to introducing *AI literacy across curriculum*, e.g., it had a powerful super-computer, and faculty members were already engaging with AI in their research, collaboration with industry, and teaching (Southworth et al., 2023).

Infrastructure Readiness: This refers to the availability and accessibility of necessary hardware, software solutions, and IT support within the institution or faculty.

Human resource Readiness: This involves assessing educators' existing AI competency, their previous experience of integrating AI into pedagogy, their familiarity with AI-driven course development or research. Given the historical imbalance between investments in technological infrastructure and professional development, low AI competency levels among teacher educators are anticipated.

Situation analysis can be used to conduct a baseline assessment and collect comprehensive evidence about needs, gaps, deficiencies, limitations and affordances. Methods such as document analysis, surveys, questionnaires, and semi-structured interviews with institutional leaders, professors and teacher educators, and IT personnel can ensure an evidence-based, participatory approach, where multi-stakeholders needs and perspectives are included. In the meantime, it will enhance the alignment between the program goals and current level of digital and AI preparedness.

4.2 B. Specifying AI literacy competencies

After assessing the AI readiness, the next step entails specifying the core AI competencies that teacher educators should develop to meaningfully leverage AI-technologies in educational contexts. These competencies serve as foundational framework, shaping both the content (what should be taught) and the expected learning outcomes (mastery levels).

Building on existing frameworks (e.g., Ng et al., 2021; UNESCO, 2024), we propose that an AI literacy program for teacher educators should encompass three core competencies: (I) foundational knowledge of AI, (II) application or AI-enhanced teaching, and (III) evaluation and AI ethics. Below, we outline the key concepts which should be taught and learned for each of these competency areas.

While a large proportion of the curriculum content and topics will be general-purpose, it will allow for differentiation to accommodate discipline-specific needs and varying levels of learners (e.g., early childhood vs. secondary education). For instance, a kindergarten teacher requires entirely different applications, tools and approaches than a high school mathematics teacher.

(I) Foundational knowledge of AI

This component or module provides teacher educators with a basic understanding of AI. Topics to be covered may involve, but not limited to: definitions and types of AI (narrow AI, general AI, super-intelligence), differences between human and machine intelligence, • The history and evolution of AI, AI subfields or domains (machine learning, natural language processing, computer vision), big data concepts (e.g., 5Vs of data, data-algorithm relationship, data quality), knowledge representation, machine learning techniques (supervised, unsupervised, reinforcement learning, artificial neural networks, deep learning), natural language processing (e.g., LLMs), applications of AI in education (intelligent tutoring systems, adaptive learning systems, learning analytics, automated scoring, eye-tracking, pedagogical games) (Caspary-Sadeghi, 2025, 2026; Lee et al., 2022; Richter et al., 2019).

All educators, regardless of subject or grade level they teach, should acquire the basic knowledge and understanding of AI. However, advanced AI literacy—particularly for educators teaching future AI professionals—may require additional training in mathematics, statistics, and data science. For example, foundational knowledge in linear algebra, probability, and calculus can facilitate deep understanding of fundamental concepts of AI such as complex data representation, neural networks, classification and clustering algorithms.

(II) AI Application or AI-enhanced teaching

This component equips teacher educators with practical skills and pedagogical strategies to integrate AI tools into their subject-specific instruction effectively. Key topics include working with tools, software solutions, and GenAI platforms which support the following activities:

- *Content generation:* using AI tools to produce instructional videos, digital textbooks, flashcards, illustrations, quizzes, and tests.
- *Teacher support:* utilizing AI-powered learning analytics dashboard, early warning systems, automated grading, real-time feedback, automated grading, lesson plan generation, and differentiated learning activities (games, projects).
- *Students' support:* implementing AI tools such as intelligent tutoring systems, adaptive learning systems, chatbots, automated feedback on writing, pedagogical games, and brainstorming for a project.

Several studies highlighted the significance of *prompt engineering* as a key skill in AI literacy programs (Chiu et al., 2024; Mills et al., 2024). Prompting refers to crafting an effective instruction,

i.e., by providing contextual information, rephrasing query, assigning a role to AI, or changing style, to get optimal output from GenAI platforms like ChatGPT, Gemini, DAL-E, Perplexity, Grok, and others. Educators should also learn that effective prompting require experimentation, clarity, and patience. The selection of AI tools or software should be pedagogically-driven to match both subject-matter (e.g., math vs. literature) and educational levels (e.g., primary vs. secondary education). Importantly, educators should adopt a reflective stance toward AI tools—questioning how they support engagement, collaboration, retention, creativity, problem-solving, teamwork, and overall learning outcomes.

(III) Evaluation & AI ethics

To avoid reducing educators to passive consumers of AI tools and applications, it's essential to raise critical AI literacy—awareness about the risk of overreliance on AI by developing teachers' competence in critical judgement of AI. This requires proficiency in two interrelated areas: AI evaluation and AI ethics. Both aspects overlap with each other, demanding *deliberate reflection* as well as centring *justice* in using and evaluating AI, especially to mitigate the potential risks or harm for and historically marginalized or systematically excluded groups (White and Scott, 2024).

4.2.1 AI evaluation

Educators should learn to critically assess AI-generated content instead of taking AI outputs at the face value. AI evaluation literacy enables educators to critically reflect on AI outputs by understanding AI limitations such as hallucination (i.e., when AI confidently produces false or non-existing answers which seem to be correct or logical at the first sight), context insensitivity (i.e., inability of AI systems to be conscious or understand context), and the role of training dataset in algorithmic accuracy. Educators should use this critical literacy to validate or verify AI-generated content through fact-checking or cross-referencing with authoritative sources (Peng et al., 2023; Stahl and Eke, 2024).

4.2.2 AI ethics

This competency addresses ethical and societal implications of AI technologies. Core topics include: fairness and algorithmic bias, transparency and explainability (XAI), privacy, data protection, AI's impact on agency, autonomy, and well-being, environmental and societal effects, and malicious uses of AI (UNESCO, 2023; White and Scott, 2024; Williams et al., 2023).

4.3 Implementation phase

A central consideration in this phase is the *instructor* or *teacher* who will implement this curriculum. According to UNESCO (2024), only few countries introduced AI literacy into their education systems. This implies a shortage of qualified educators in this field. To address this gap, we propose an efficient, agile, and scalable solution: “*train-the-trainer*” model. In this model, a small group of highly motivated educators, who already had

experience or knowledge in AI, receive intensive training and professional development. These trained educators then take on leadership roles within their faculties, supporting the capacity-building of their colleagues or in-service teachers in schools.

The effectiveness of curriculum implementation can be optimized through *constructive alignment*, a principle that ensures coherence among learning objectives (core competencies), pedagogy, and assessment. We suggest the following alignments during implementation: (a) aligning *delivery modes* with participants' needs and preferences, (b) aligning pedagogical approaches and instructional strategies with target proficiency areas, and (c) aligning instructional materials with specific settings or learning environments.

4.4 A. Implementation: delivery modes

To accommodate diverse needs of educators and enhance inclusivity, the AI literacy program can be delivered through multiple delivery modes, broadly categorized into: (a) formal learning, and (b) informal learning, and (c) just-in-time support (Folkestad, 2006; Long et al., 2021).

(a) Formal learning involves structured, well-prepared, time-bound courses with clear learning objectives. Such courses can be offered as discrete or stand-alone courses (Foundation of AI), integrated or merged with disciplinary-based courses (AI-enhanced mathematics teaching). Formal courses can be offered in different ways to enhance inclusivity such as classroom-based, online, hybrid, MOOCs, micro-credentials.

(b) Informal learning: in addition to formal credit-bearing courses, informal and extra-curricular learning opportunities play a vital role in developing AI literacy (Long et al., 2021), e.g., Communities of Practices (CoPs), both online and offline, where educators can learn from each other by sharing their experiences of practical examples and engage in peer learning. CoPs support reflective dialogue and collaborative problem-solving, reducing the gap between AI theory and classroom practice. Other informal learning opportunities include clubs and Special Interest Groups (SIG), brown-bag sessions, reading groups, mentoring networks, and visits to educational exhibitions. These informal learning activities should be offered both as subject-specific (e.g., AI in mathematics teaching) and cross-disciplinary (e.g., AI applications in STEM and humanities).

(c) Just-in-time support: Educators and faculty member often need immediate, context-specific support when encountering technical or pedagogical problem related to AI integration. Institution should ensure on-demand help desk or coaching.

4.5 B. Implementation: pedagogical approaches

Effectively integrating AI literacy into educational contexts require deliberate selection of instructional activities and learning experiences aligned with different variables, i.e., learning environments, participants' needs, and the specific AI competency being mastered. Below, we outline recommended

pedagogical approaches for teaching each of the three core AI literacy competencies:

1. Teaching AI Foundation: using lectures, group discussion and reflection, visualizing strategies such as concept map or illustration.
2. Teaching AI use and applications: using hands-on, experiential learning, and active involvement of the participants, such as project-based learning, problem-based learning, inquiry-based learning, and design of a chatbot or game.
3. Teaching AI Evaluation and Ethics: using case study, scenario-based learning, storytelling, and role-play using concrete and tangible real-world examples. However, educators should not bear sole responsibility for AI tool safety; institutions or local authorities must ensure compliance with ethical and safety standards.

4.6 C. Implementation: aligning materials to setting

When designing AI literacy materials and content, it's crucial to consider different instructional environments or settings to ensure effectiveness and accessibility. Below, we suggest various materials and resources based on two types of environments: low-tech and AI-rich settings.

Unplugged settings: refer to offline environments which include low-tech solutions and low-cost resources. These settings can incorporate paper-based resources such as printed articles, book chapters, worksheets, flipcharts, posters, scripts, reports, and checklist. Additionally, pre-downloaded or recorded videos or podcasts can facilitate teaching and learning of foundational AI concepts or AI evaluation and ethics.

Online settings: refers to fully-connected, digital environments and AI-based technologies to deliver multimodal resources and materials. Resources such as mobile applications, generative AI platforms and chatbots, online games, simulations, virtual labs, digital books, social media, virtual and augmented reality tools (VR/AR) technologies and other available AI tools and software.

4.7 Program evaluation

To establishing the effectiveness, relevance, and utility of AI literacy program in practice, it's essential to conduct a rigorous program evaluation. Rather than relying merely on a summative evaluation, we prioritize participatory, system-wide, and inquiry-based evaluation models. Patton (2003, 2008) suggests a utilization-focused evaluation, where the primary goal of evaluation is to enhance the practical use and uptake of findings by intended-users, such as educators, institutional leaders, and instructional designers.

In addition to involving intended users early and throughout the entire program design and implementation process, an

inquiry-based approach is suggested by collecting evidence through various types of evaluation, including:

1. *Summative* evaluation, which measures program outcomes and achievements at the end of the program,
2. *Formative* evaluation, which collects regular feedback from participants for continuous improvement during the program implementation,
3. Process and developmental evaluation, which collects evidence about what works and what doesn't work in the process of program's implementation.
4. Outcome evaluation, which examines the impacts of program.

5 Challenges and strategies

The proposed AI literacy framework offers a structured approach for preparing teacher educators to integrate AI meaningfully into their practice. However, as a conceptual model, its implementation is subject to several limitations and challenges that we discuss below and suggests strategies to address them.

1. *Human Resources:* the primary challenge is ensuring a qualified workforce, including educators, evaluators, module developers, and coaches. AI literacy requires sustained development, not short-term workshops. Most institutions are only beginning to integrate AI literacy, and comprehensive curricula for teacher educators remain limited. A recurring issue is educators' limited background in statistics and mathematics, which are essential for understanding AI fundamentals and evaluating outputs (Caspary-Sadeghi, 2026; Mandinach and Abrams, 2022). Without this foundation, AI literacy becomes superficial. A viable strategy is agile program prototyping, starting with a "train-the-trainer" model to foster educators' confidence, gradually increasing complexity for deeper conceptual understanding.
2. *Keeping up with AI:* AI evolves rapidly, making it difficult to keep curricula, tools, and ethical guidelines current. What counted as AI a decade ago may now be obsolete. Distinguishing human- from machine-generated content is also increasingly complex. A solution is modular curricula that can be easily updated without replacing the entire program.
3. *Lack of Users' Engagement:* The effectiveness of the proposed framework relies heavily on active participation from teacher educators throughout design and implementation. However, engagement goes beyond passive feedback collection and requires genuine collaboration, reflective practice, and shared ownership. Consistent with research on educational change (Fullan, 2015; Maxwell, 2021), challenges may include limited time, competing professional demands, or insufficient incentives for innovation. The strategies proposed to mitigate them such as allocating dedicated time for AI learning, strengthening professional learning

communities, incentivizing experimentation, and promoting practitioner research, are promising but depend on organizational support and cultural readiness. Without these enabling conditions, engagement and the framework's impact will remain constrained.

4. *Infrastructure and Access Inequities*: A further limitation relates to the uneven availability of technical infrastructure necessary for sustained AI literacy development. Implementing the proposed framework necessitates reliable access to computational tools, updated software, secure data environments, and adequate technical support. Moreover, concerns related to data privacy, cybersecurity, and institutional policies on AI tool usage may further constrain implementation.
5. *Lack of evidence-based best practices*: The final barrier stems from the limited availability of robust empirical research on effective pedagogical solutions. Particularly, there is insufficient evidence on which teaching approaches, instructional strategies, and curriculum designs in higher education best support the progression from basic AI understanding to applied, discipline-specific AI competencies. Additionally, little is known about how AI competency development varies across academic disciplines and among students with differing educational backgrounds, levels of digital proficiency, and professional trajectories.

To summarize, these challenges are not isolated but reflect broader systemic, institute, and teacher-level barriers that collectively constrain AI integration in education. These barriers include personal, technological, and organizational factors, e.g., limited teacher preparedness, low technological self-efficacy, insufficient institutional support, persistent misconceptions about AI, lack of training and resources, and concerns regarding bias and accuracy, which together highlight the need for coordinated structural and professional development strategies (Ahmed et al., 2024; Othman et al., 2024).

6 Conclusion

As AI continues to impact educational practices, AI literacy is emerging as a foundational competency for educators. This conceptual article has argued that *teacher educators*—those who prepare the next generation of teachers—require more than surface-level familiarity with AI tools. The conceptual study has addressed critical gaps in the development of comprehensive curricula for teacher educators' professional learning in AI literacy. Building on established literature, empirical research, and the author's decade of experience in designing faculty development programs across international contexts, this study proposed a structured and contextually adaptable curriculum for developing AI literacy among teacher educators in higher education based on the elements of BDR. Rather than prescribing a fixed sequence or hierarchical steps, the framework is intentionally designed as a flexible blueprint—an open-ended roadmap that can be adapted to diverse institutional landscapes.

By incorporating a holistic (e.g., formal, informal, and just-in-time learning mechanisms) and evidence-based approach

(rigorous, utilization-focused assessment), the framework ensures the sustainability and long-term impact of AI literacy program. Furthermore, through a participatory co-design process, the curriculum aligns with institutional needs while equipping educators with the required skills to critically engage with AI in educational contexts. We also problematized its implementation by identifying potential challenges and suggesting strategies to address them. Future research should focus on empirically validating the framework by evaluating its effectiveness across diverse educational contexts, examining its impact on educators' cognitive, behavioural and motivational factors, and refining strategies for AI integration into different learning environments. Further relevant research directions include development of rigorous benchmarks and standardized metrics for assessing AI competency development in teachers. Additionally, longitudinal studies are needed to explore the long-term effects of using AI on teacher professional development, particularly in terms of their creativity and ability to engage in life-long learning independently.

Author contributions

SC-S: Writing – original draft, Writing – review & editing.

Funding

The author declares that financial support was not received for this work and/or its publication.

Conflict of interest

The author declares that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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